

Frame Members — Front Frame Horn



Special Tool(s)

 ST3049-A	Rust Inhibitor Installation Kit 286-00002
 ST3048-A	Undercoating Spray Gun 286-00001

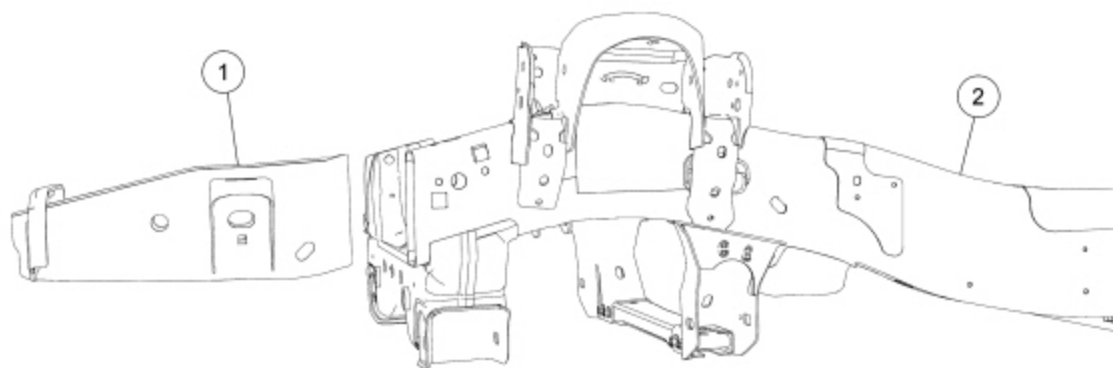
General Equipment

3 Phase Inverter Spot Welder 254-00002
Compuspot 700F Welder 190-50080
I4 Inverter Spot Welder 254-000014
Inverter Welder with MIG Welder 254-00015

Material

Item	Specification
Motorcraft® Metal Surface Prep ZC-31-B	—
Premium Undercoating ValuGard™ VG101, VG101A (aerosol)	—
Rust Inhibitor ValuGard™ VG104, VG104A (aerosol)	—

Front Frame Horn



N0096296

Item	Part Number	Description
1	5D059 5D059 LH/ 5D058 RH LH/ 5D058 RH	Frame rail replacement kit
2	5005 5005	Frame assembly

WARNING: Collision damage repair must conform to the instructions contained in this workshop manual. Replacement components must be new, genuine Ford Motor Company parts. Recycled, salvaged, aftermarket or reconditioned parts (including body parts, wheels or safety restraint components) are not authorized by Ford.

Departure from the instructions provided in this manual, including alternate repair methods or the use of substitute components, risks compromising crash safety. Failure to follow these instructions may adversely affect structural integrity and crash safety performance, which could result in serious personal injury to vehicle occupants in a crash.

WARNING: Frame rail crush zones absorb crash energy during a collision and must be replaced if damaged. Prior to replacement of frame rail crush zones, straighten damaged frame rails to correct frame dimensions. Failure to follow these instructions may adversely affect frame rail crush zone performance and may result in serious personal injury to vehicle occupants in a crash.

WARNING: Invisible ultraviolet and infrared rays emitted in welding can injure unprotected eyes and skin. Always use protection such as a welder's helmet with dark-colored filter lenses of the correct density. Electric welding will produce intense radiation, therefore, filter plate lenses of the deepest shade providing adequate visibility are recommended. It is strongly recommended that persons working in the weld area wear flash safety goggles. Also wear protective clothing. Failure to follow these instructions may result in serious personal injury.

WARNING: Always wear protective equipment including eye protection with side shields, and a dust mask when sanding or grinding. Failure to follow these instructions may result in serious personal injury.

NOTE: Observe prescribed welding procedures when carrying out repairs to the frame assembly. For additional information, refer to Welding Precautions — Steel in this section.

- NOTE:** All body alignment measurements are carried out with the vehicle detrimmed. Measurements are made metal to metal, on center, unless otherwise specified.

Measure the vehicle to determine if the body requires straightening and alignment. The vehicle must be restored its correct overall dimensions prior to beginning this procedure. For body dimensional information, refer to Body in this section.

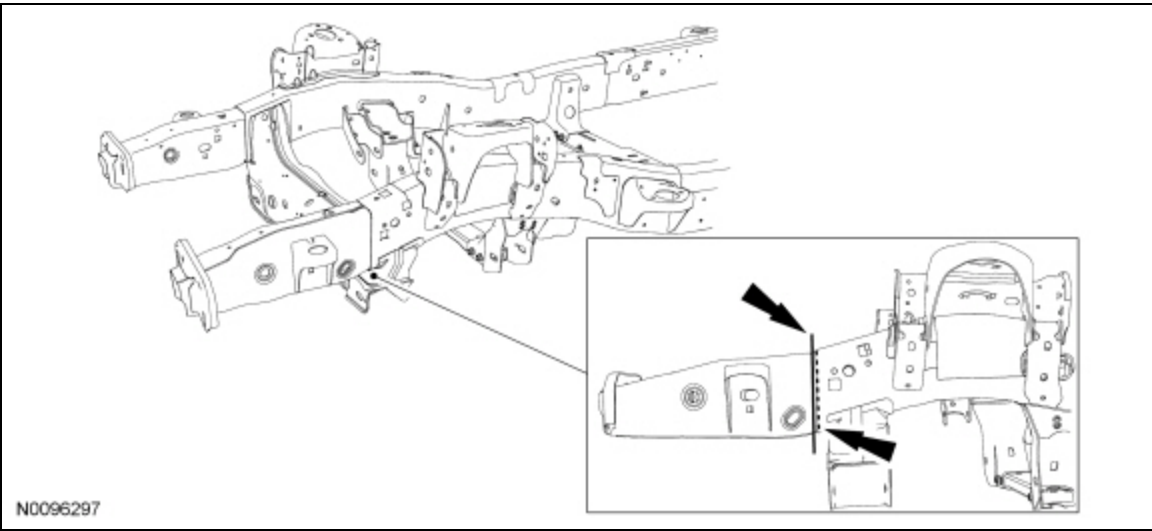
- Anchor the vehicle to a frame rack following the equipment manufacturer's instructions.

- Remove the bumper. For additional information, refer to Section 501-19.

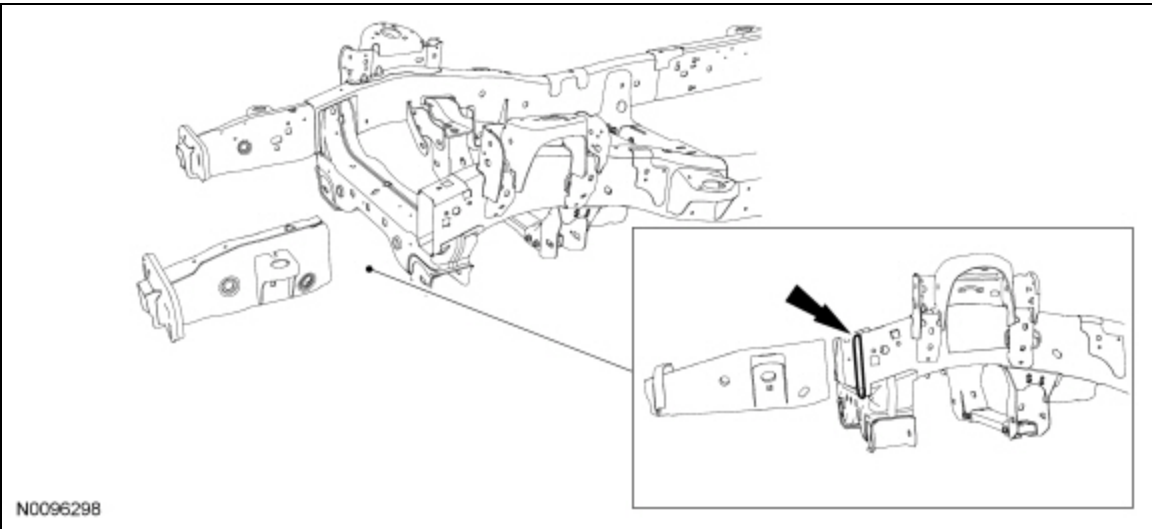
- Unbolt the Front End Sheet Metal (FESM) body mount bolts.

- Raise the body if required. For additional information, refer to Section 502-02.

4. Locate the vertical weld of the front frame horn to center rail.



5. Using a plasma cutter, reciprocating saw or cut-off wheel, remove the damaged front frame horn section. Do not cut directly along the weld line. Leave enough material on the front side of the weld line to allow the edge to be ground back exactly to the line. This is necessary to make sure of correct fit between the frame and the replacement frame section.
6. Grind off the excess welding material that remains in front of the weld line on the existing frame.
- Remove the remaining piece (ring-like) of the front frame horn from inside the mid-rail.
7. Using a wire brush or sandpaper, remove the E-coat from the outer surfaces of the replacement service section within approximately 15 mm (0.59 in) of the repair joint.
- Using a wire brush, remove any foreign material from the frame within approximately 15 mm (0.59 in) of the repair joint.



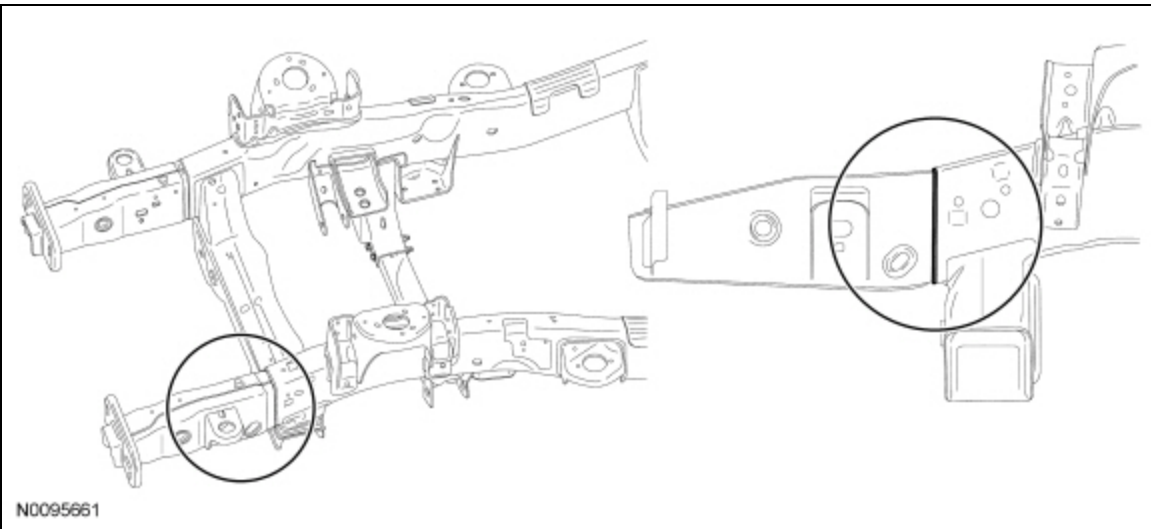
8. **NOTE:** The front frame rail replacement section is tapered at the ends to fit into the frame.

NOTE: The frame rail service component must be located to maintain original factory dimensions. For additional information, refer to Body in this section for correct frame dimensional information.

Install the replacement front frame horn section inside the center rail of the frame. Support the replacement service section in position. Check that the FESM body mount bracket lines up with the sheet metal hole for the body mount.

9. Make sure the repair joint and surrounding repair area have attained a minimum temperature of 10°C (50°F) before carrying out the weld procedure.
10. Tack weld at the 4 corners and check the fit and alignment.
- Lower the body onto the frame and recheck fit and alignment, then raise it.

11. Weld the joint completely around the perimeter, if fit and alignment are correct, using a Metal Inert Gas (MIG) welding machine capable of producing a minimum of 200 amps. Use 0.9-1.1 mm (0.035-0.045 in) ER70S-3 or equivalent weld wire that is compatible with mild (Society of Automotive Engineers (SAE) 1010) steel.



12. Dress the weld(s) as necessary.
13. Use a dye penetrant to determine if any cracks or large voids exist in the weld joint. If cracks or other defects exist, grind out the defect and repair until the weld is free of defects.
- Clean the repair area with metal surface prep and apply anti-corrosion coating to the affected areas of the frame and service replacement section. For additional information regarding corrosion protection, refer to Restoring Corrosion Protection Following Repair in this section.
14. Reinstall components removed during disassembly. Make sure all fasteners are tightened to the correct specification.